

08. reading and writing files

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Learning objectives

- Be able to open, read and write files
- Navigate folders and manipulate paths

Navigating folders

- Sometimes we need to work with many files at once, like reading all the files in a folder.
- Python has a built-in module called `os` that helps us look at and navigate the files and folders on our computer.
- With `os`, we can:
 - See which files and folders are in a folder
 - Check if a file or folder exists
 - Create new folders
 - Change the current working folder
- This lets us write programs that work with multiple files automatically, instead of opening them one by one.

Navigating folders

```
import os

# get path to the folder we're currently in
os.getcwd()

# see contents of the current folder
os.listdir()

# change to a new directory
os.chdir()
```

Working in Google Colab

- When we work in Google Colab, our code runs on a virtual machine in the cloud
 - Do not automatically see your files on your Google Drive
 - It only has access to its own temporary storage (called `content`)
- We need to mount of Drive to “**connect our Google Drive to the virtual computer**”. Once we do this, it has access to our drive !

```
# Step 1. Import the drive module
from google.colab import drive
# Step 2. Mount it
drive.mount("/content/drive")
# Step 3. Set working directory
os.chdir("/content/drive/MyDrive/")
```

Navigating folders

```
# Path to the folder you want to look inside
folder_path = '01_material/data'

# Get a list of all files and folders inside
items = os.listdir(folder_path)

# Print each item
for item in items:
    print(item)

delay_reasons.csv
all_delays.csv
bike_thefts_joined.csv
california_housing_test.csv
...
```

Opening, reading and writing files

- Python can read and write files using the built-in `open()` function.
- Syntax: `open(file_path, mode)`
 - `file_path`: string indicating the location of the file
 - `mode`: one-character string indicating how to open the file

Mode	Description
'r'	Read only. Produces an error if the file doesn't exist
'w'	Write. Creates a new file if it doesn't exist, or overwrites an existing file
'a'	Append. Adds to an existing file, or creates a new file if it doesn't already exist

Opening files

- To open a file, we use a with statement which follows the pattern
 - `with open('file path') as variable_name`
- Then an indented block of code to process the file

```
# Example
with open '01_material/data/
california_housing_test.csv', 'r') as file:
    print(file)
```

This tells you Python successfully opened the file.

```
<_io.TextIOWrapper
name='../..05_src/data/slides_data/california_housi
ng_test.csv' mode='r' encoding='UTF-8'>
```


Reading files

- As we saw, opening a file doesn't immediately get us the file's contents. To do that, we must use a read method

Method	Description
<code>read()</code>	returns the full file contents, which can be overwhelming for larger files
<code>readline()</code>	returns only the next line in the file. Python keeps track of where it is in the file.
<code>readlines()</code>	returns the full file as a list. Each item is one line in the file.

```
with open('01_material/data/california_housing_test.csv', 'r') as file:  
    print(file.read())
```

This reads the entire file!

Reading files

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Method	Description
<code>read()</code>	returns the full file contents, which can be overwhelming for larger files
<code>readline()</code>	returns only the next line in the file. Python keeps track of where it is in the file.
<code>readlines()</code>	returns the full file as a list. Each item is one line in the file.

```
with open('01_material/data/california_housing_test.csv', 'r') as file:  
    print(file.readline()) # First line  
    print(file.readline()) # Second line
```

This reads the line one at a time

Reading files

- As we saw, opening a file doesn't immediately get us the file's contents. To do that, we must use a read method

Method	Description
<code>read()</code>	returns the full file contents, which can be overwhelming for larger files
<code>readline()</code>	returns only the next line in the file. Python keeps track of where it is in the file.
<code>readlines()</code>	returns the full file as a list. Each item is one line in the file.

```
with open('01_material/data/california_housing_test.csv', 'r') as file:
    lines = file.readlines()
for line in lines:
    print(line)
```

This reads everything in the file as a list of lines!

Writing files

- There are also methods to write.

Method	Description
<code>write()</code>	Writes a single string to file
<code>writelines()</code>	Writes a list (or any iterable) of strings to a file. Need to separate new lines

```
airlines = [  
    "Code\tAirline\n",  
    "AAL\tAmerican Airlines\n",  
    "ABK\tAirbus Canada\n",  
    "ABP\tABS Jets\n",  
    "ACA\tAir Canada\n"  
]
```

Recall the escape sequences!

```
with open('airlines.txt', 'w') as file:  
    file.writelines(airlines)
```

Code	Airline
AAL	American Airlines
ABK	Airbus Canada
ABP	ABS Jets
ACA	Air Canada

Appending files

- There are also methods to add to an existing file (append).

```
with open('airlines.txt', 'a') as file:  
    file.write('ARQ\tAirico\n')
```

Code	Airline
AAL	American Airlines
ABK	Airbus Canada
ABP	ABS Jets
ACA	Air Canada
ARQ	Airico

We added a code and airline to our existing text file!

Quiz: Reading and writing files

1. What does a file path refer to?

- A. The name of a variable
- B. The location of a file on your computer
- C. The contents of a file
- D. It's a python keyword

Quiz: Reading and writing files

2. Which of the following mode opens a file for reading only

- A. 'r'
- B. 'w'
- C. 'a'
- D. None of these

Quiz: Reading and writing files

3. Which of the following mode adds text to an existing file without deleting its contents?

- A. 'r'
- B. 'w'
- C. 'a'
- D. None of these

Quiz: Reading and writing files

4. What happens if you open a file in 'w' mode that already exists?

- A. Python raises an error
- B. The file is deleted
- C. The file is overwritten
- D. The file is opened in read-only mode

Quiz: Reading and writing files

5. What does this code do?

```
with open('file.txt', 'a') as file:  
    file.write('Hello! \n')
```

- A. Deletes the file
- B. Overwrites the file
- C. Adds Hello! to the end of the file
- D. Reads the file

Quiz: Reading and writing files

6. Complete the code so it writes “Hello, Python!” to a file called output.txt

```
with open('output.txt', 'w') as file:  
    file._____('Hello, Python!')
```

Exercise: Part 1

- Write a code that:
 1. Asks the user for their name
 2. Saves it to name.txt

Exercise: Part 2

- Write a code that:
 1. Asks the user for their age
 2. Append it to name.txt

Exercise: Part 3

- Write a code that:
 1. Open name.txt
 2. Read and print the contents